

Practical Network Defense

Notes collected during lessons. Not corrected.

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IMPORTANT DISCLAIMER: These notes have been taken during lessons. They might contain errors (specifically, grammatical errors) and be incorrect. Please always check other sources to confirm what is written is correct. I don't think I'll correct them after lessons, though it's possible.

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Chapter 1

Introduction

Here is the website for the course:

<https://sites.google.com/di.uniroma1.it/practical-network-defense>.

Chapter 2

Part 1

Internet Standards enumerates design documents that states how things work on the Internet.

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Remember the structure of the internet (hierarchical structure), given by the Backbone, then the ISPs, then organizations and LANs. ISPs, especially, can be divided into three layers, from Tier 1 to Tier 3.

We also have this hierarchy:

1. Network Edge
2. ?

The number of Tier 1 ISPs is quite limited, like Verizon. These ones are connected to IXP (Internet Exchange Points), which is like a switch where every company can plug its cables.

Who generates traffic? A LAN, typically. Usually, this LAN is able to cover only a small and localized area of hosts. The structure of a LAN is given by several hosts, connected to an edge router (gateway). This is a two-sided device, with one side being inside the LAN, and the other connected to the global Internet. A lot of protocols are used in the functioning of this mechanism. Remember that a single packet is encapsulated into different protocols. We need this encapsulation for several advantages. The upper layer protocols could always be the same, and we can change lower layers without impacting anything. We can, for example, use a TCP packet over IP, and so on.

A local host is an host which can communicate directly with local hosts. For remote hosts it's different, as we need to go through other routers to reach the other destination (so two local hosts have a direct link between them).

Ethernet frames are used in this case. The packet contains some fields like the preamble and SFD, which are used for synchronization. We also have destination and source, those contain addresses, which are called MAC addresses. They're 48 bits long. We then have the payload and the frame closure bits. How do we build an Ethernet network? Specifically, in the real world we have

several channels connected to a switch. So, we have many hosts connected to a single device, which moves packets into the Ethernet. The idea is that every single packet inserted into an Ethernet network can reach any other host in the network domain, that's why we call this the broadcast domain. ARP, for example, uses broadcast messages.

Switches, however, do not always broadcast. Rather, it sends the packet on the link associated to the ethernet destination field addressed. Why Internet is not a large Ethernet network? As we make a lot of use of broadcasts, which could not be suitable. We need to distinguish between a logical division of the network, and a physical one (for IP).

Part missing.

2.1 Practical Part